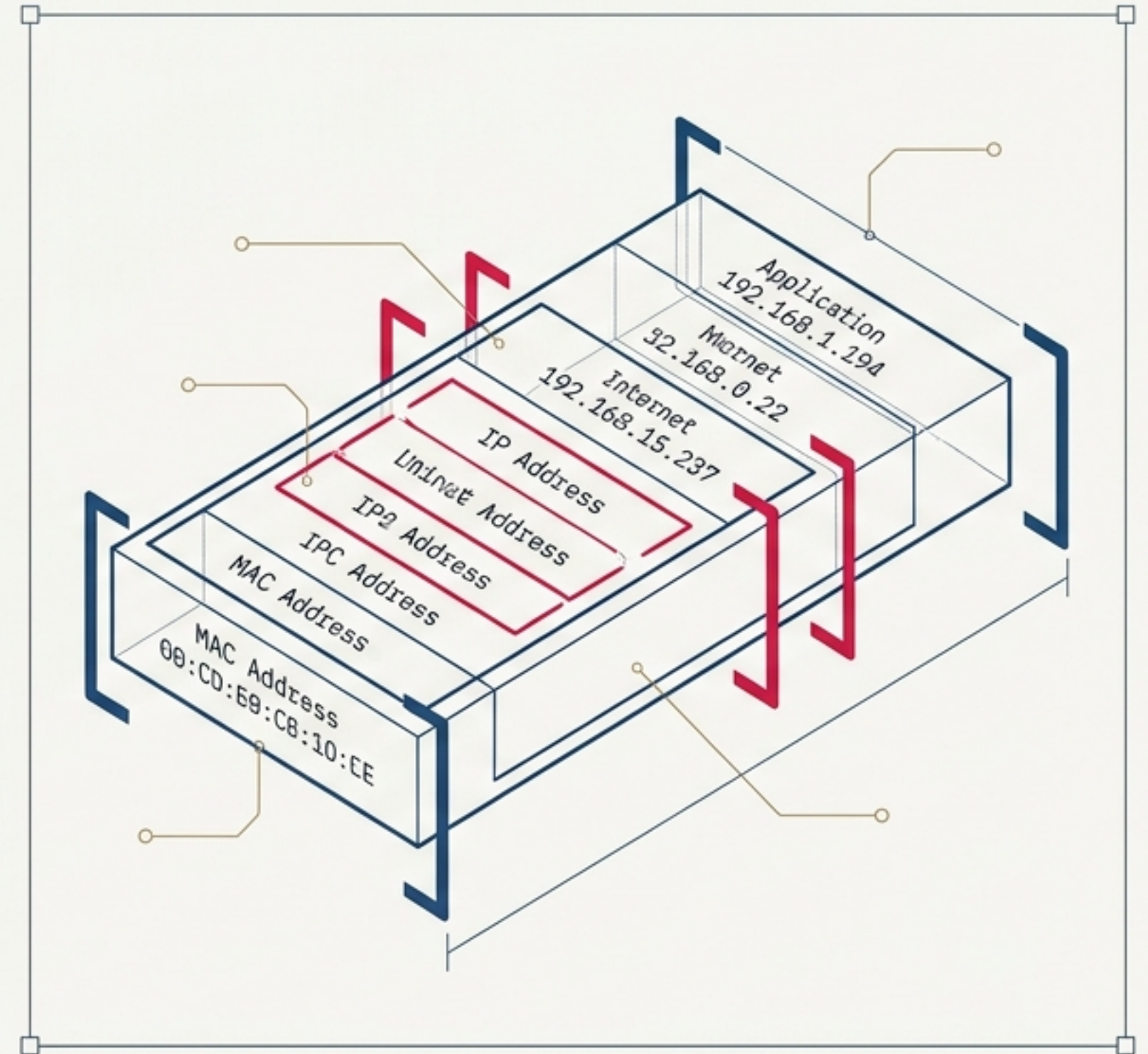


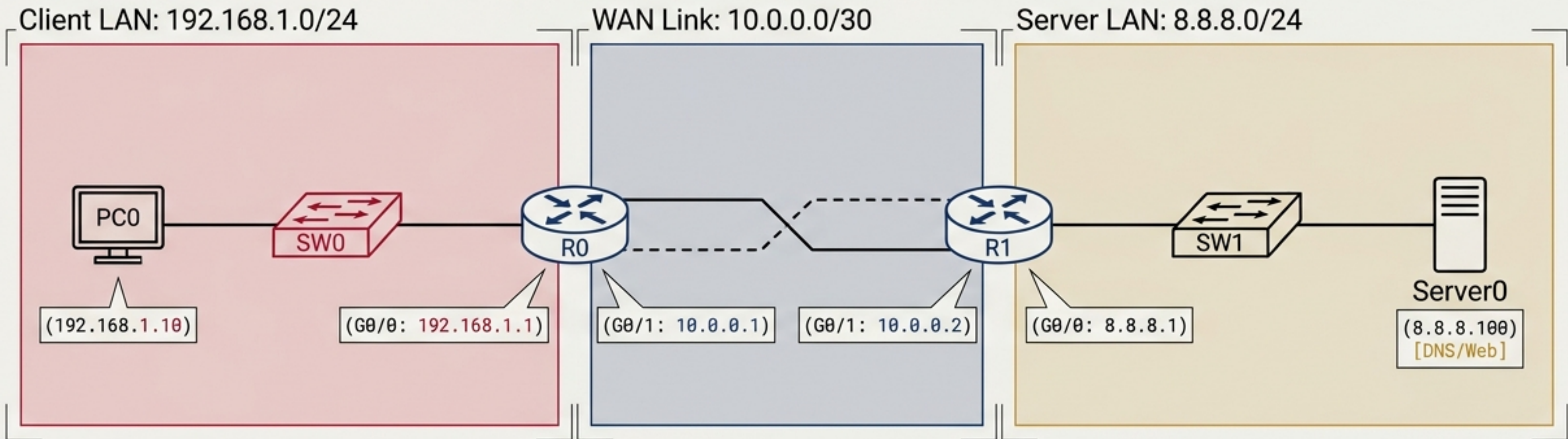
Network Forensics: Blueprint of a Web Request

Visualizing encapsulation, routing, and data flow from the physical layer to the application layer.

SOURCE: Packet Tracer Simulation Data



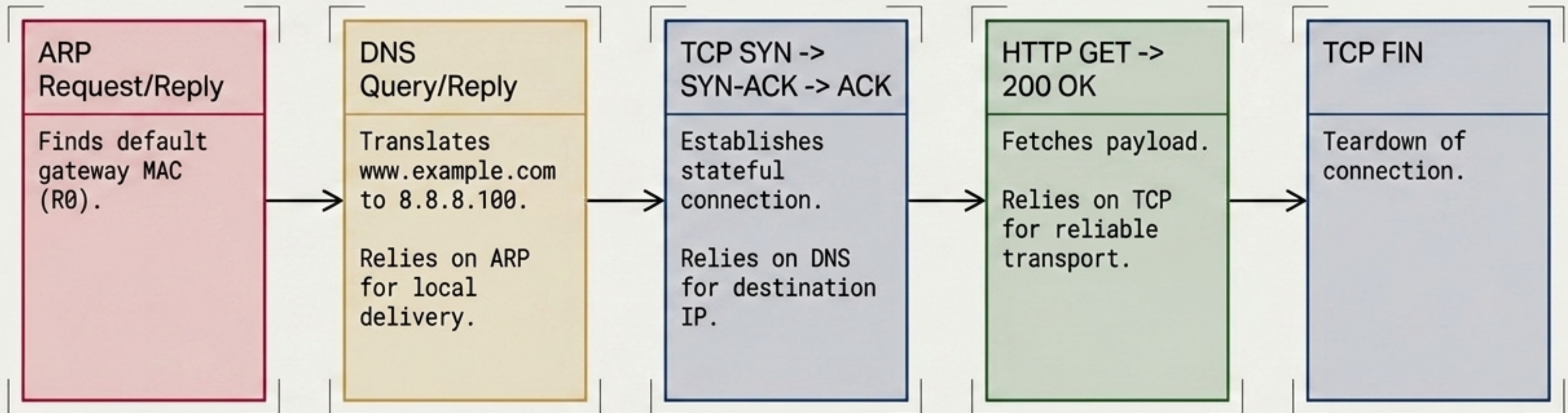
The Topography of a Web Request



The Routing Mandate:

PC0 cannot reach Server0 directly because they reside on different logical subnets. R0 and R1 must rebuild the connection across the WAN.

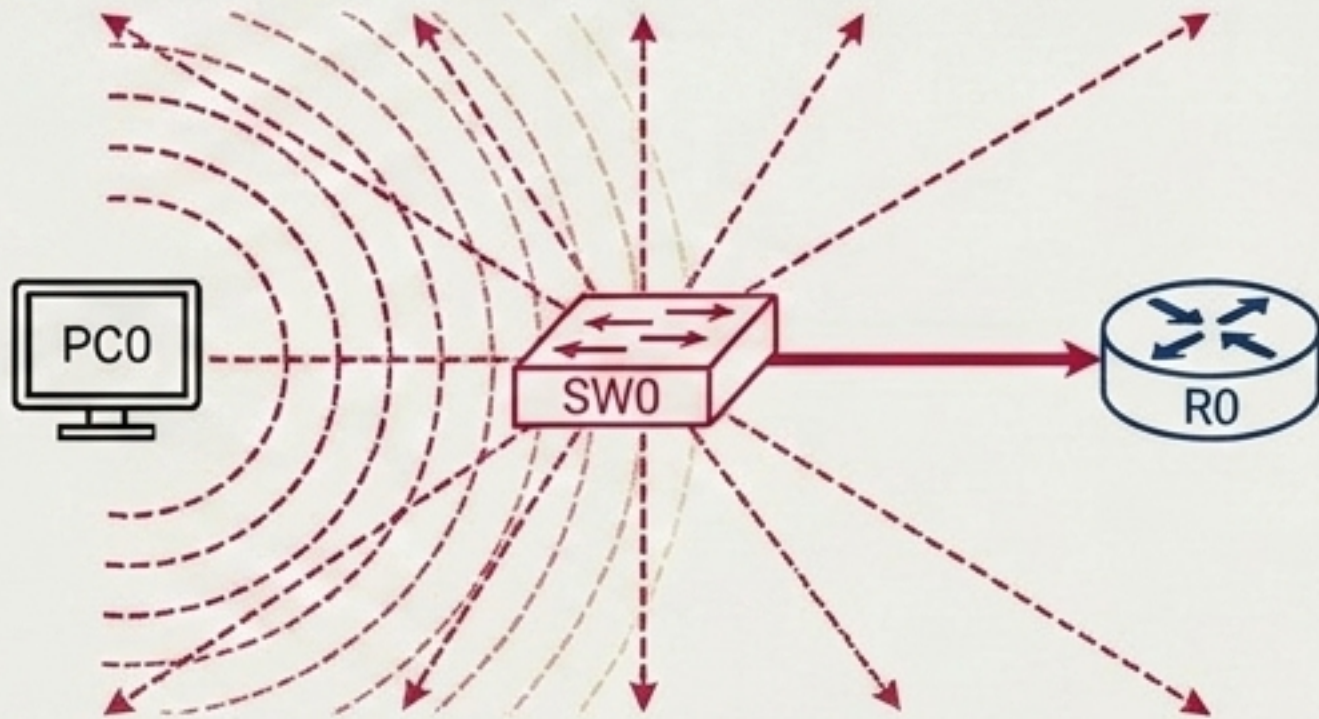
The Chronology of a Single Click



Failure at any stage halts the entire sequence.
If DNS fails, the TCP handshake never begins.

Resolving the First Hop: ARP

PC0 knows R0's IP (192.168.1.1) from its default gateway settings, but needs its physical address.



Dest MAC: **FF:FF:FF:FF:FF:FF**
ARP Target IP: 192.168.1.1

Sent to all switch ports (Broadcast).

R0 identifies its own IP and responds directly.

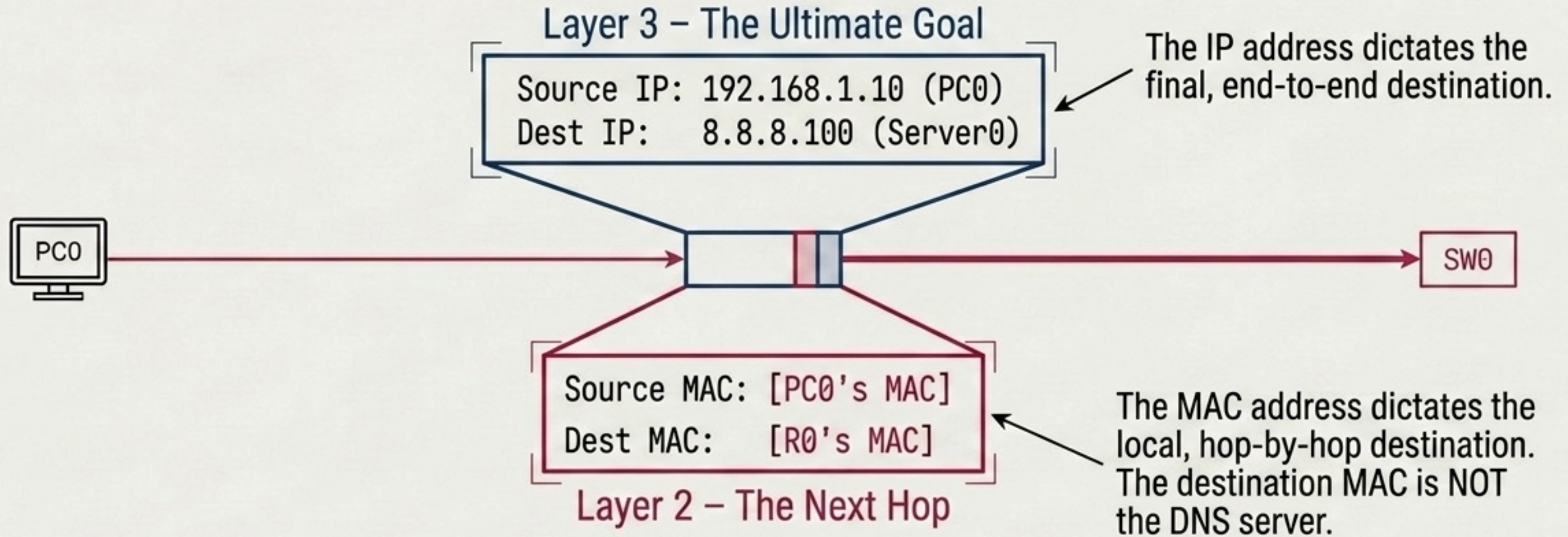


Dest MAC: **[PC0's MAC]**
ARP Sender MAC: **[R0's MAC]**

Unicast. PC0's ARP cache is now populated.

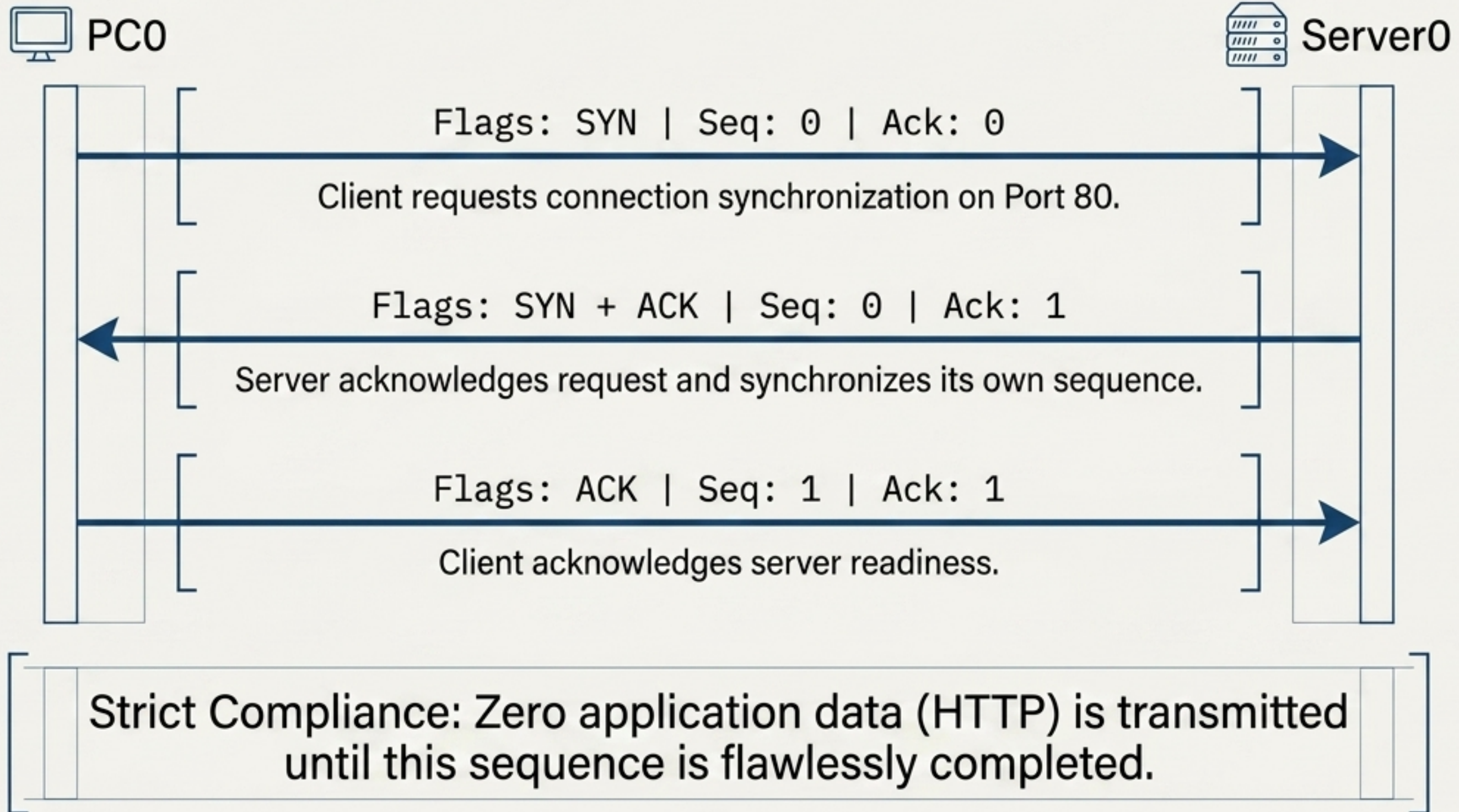
The Split Destination Dilemma

Target: DNS Query UDP Packet (Port 53)

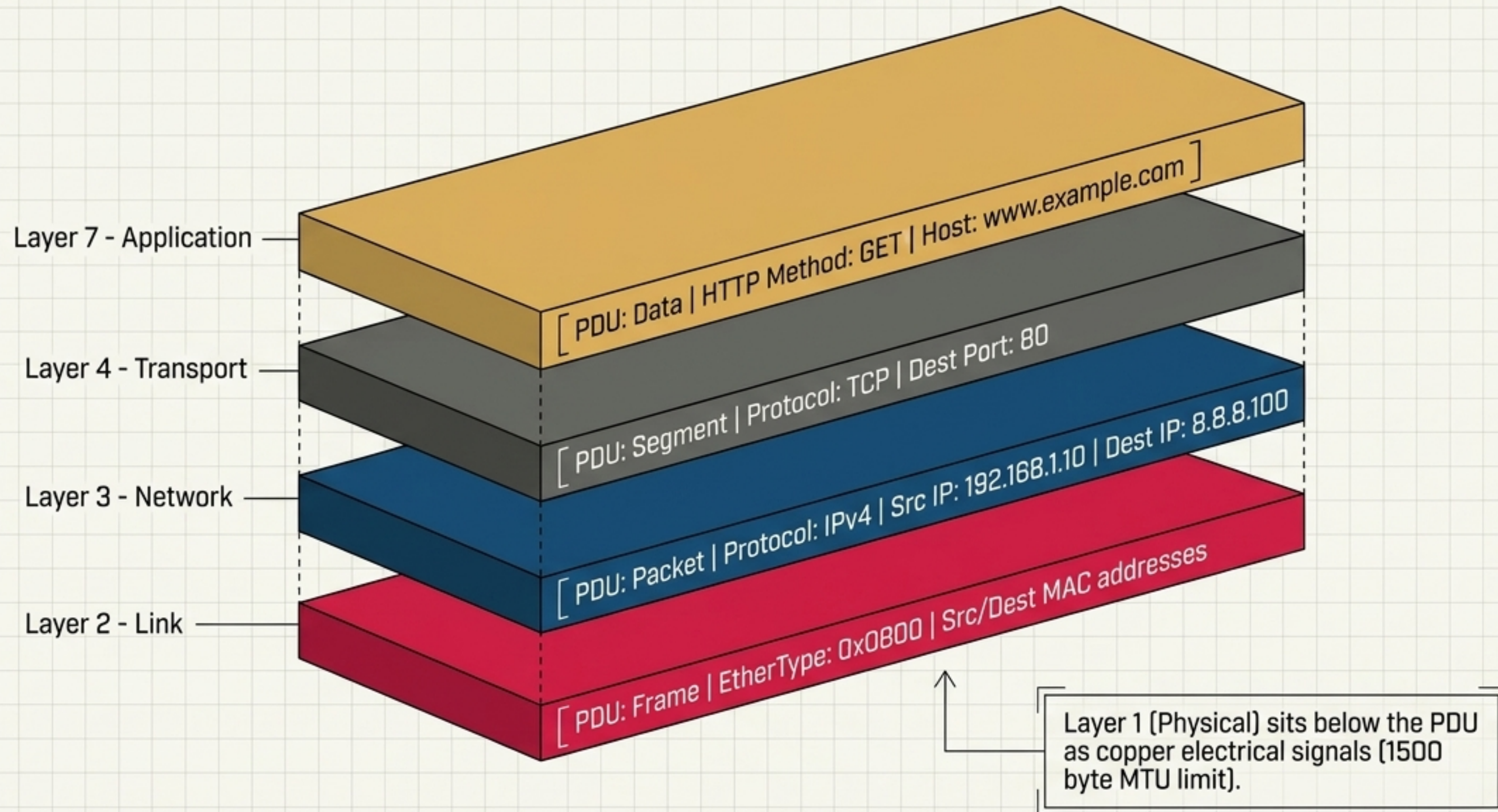


Because 8.8.8.100 is on a different subnet, PC0 hands the frame to its default gateway (R0).

Initiating State: The TCP Handshake

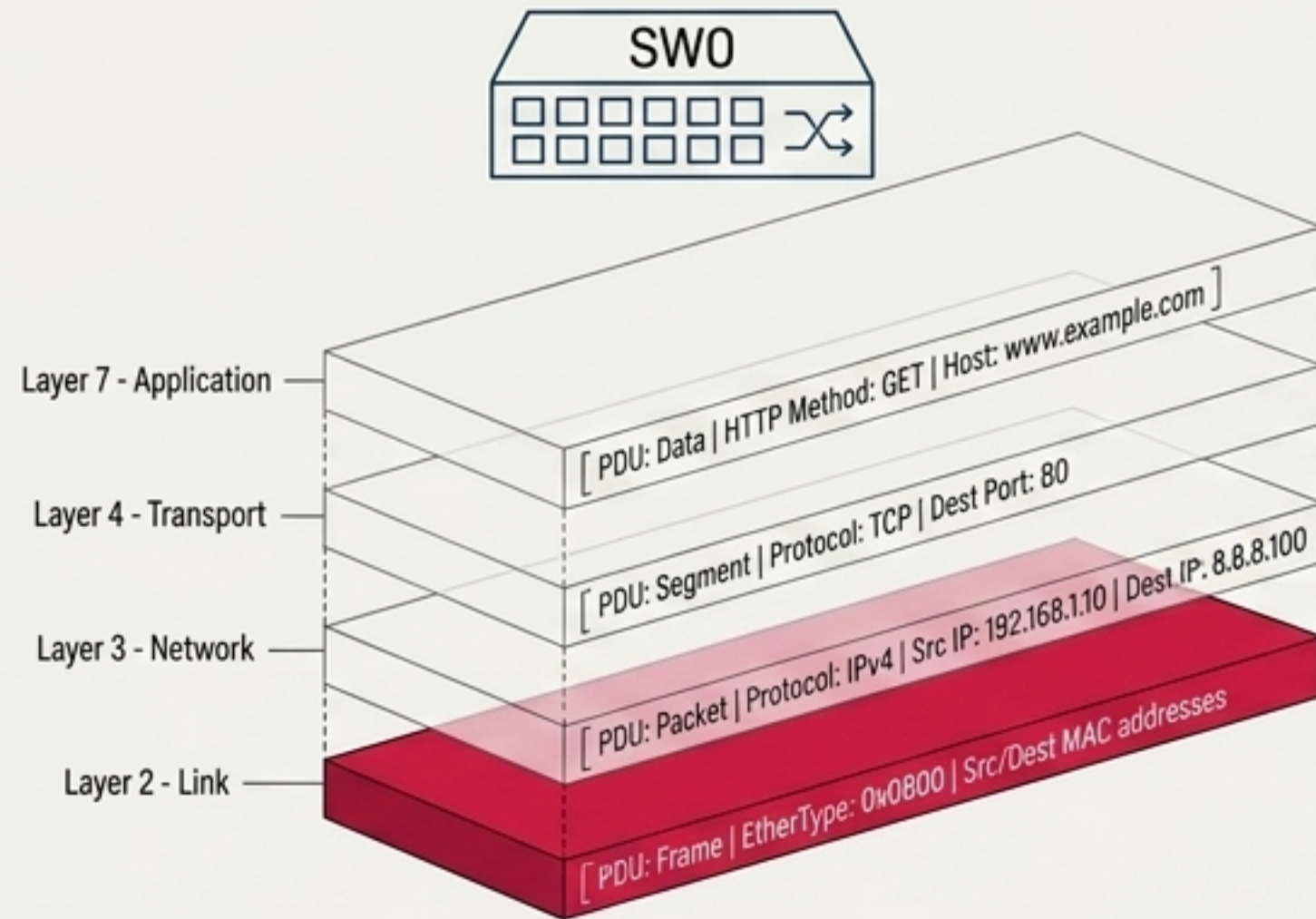


The Architecture of Encapsulation



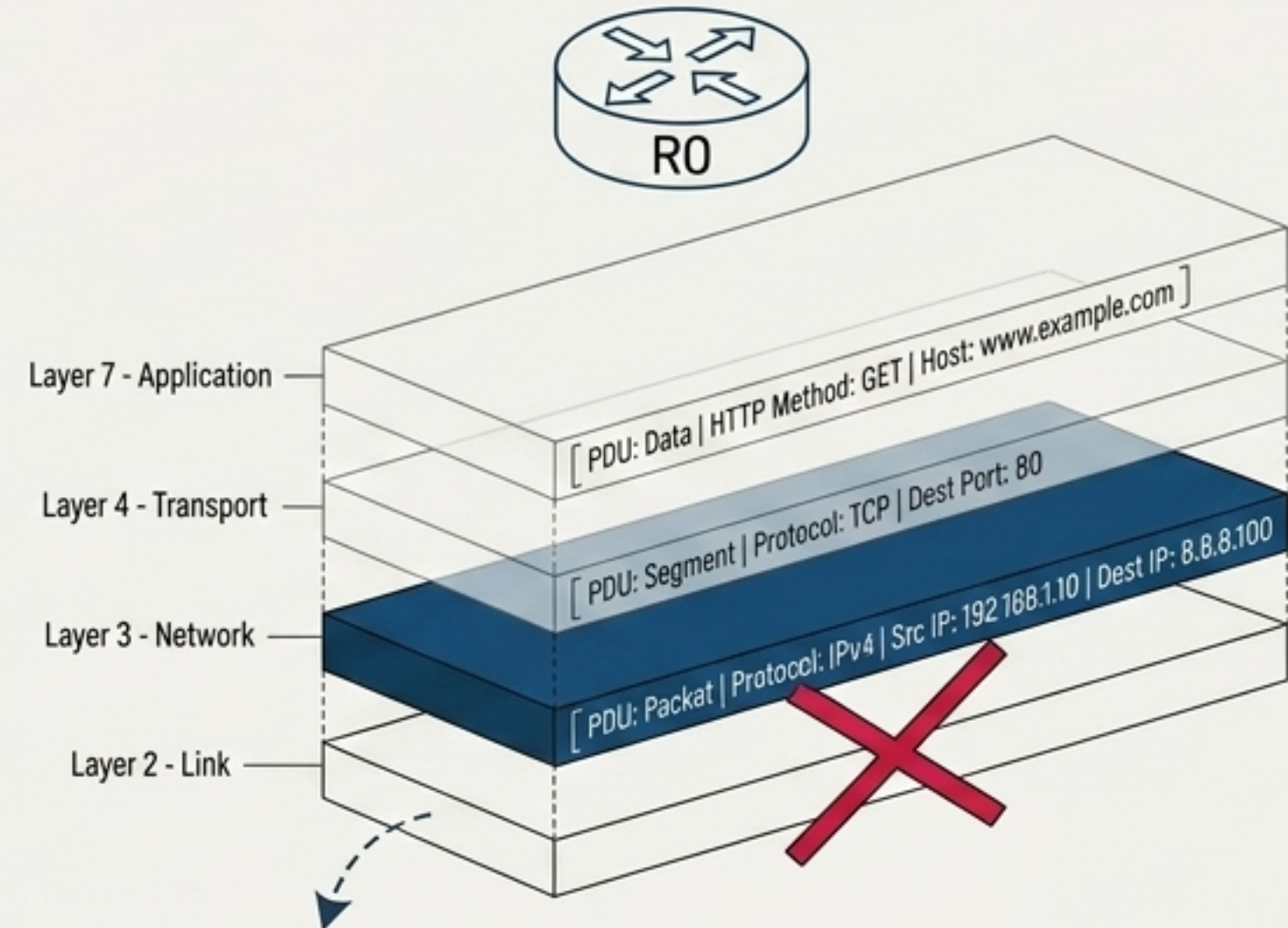
Layer Independence & Separation of Concerns

The Switch (SW0)



Does SW0 look at IP or TCP? No.
Switches make forwarding decisions based strictly on the Ethernet destination MAC.

The Router (R0)



Does R0 look at the TCP ports or HTTP payload? No.
Routers strip the old frame, read the IP destination, and build a completely new frame for the next hop.

The Hop-by-Hop Matrix: Frame vs. Packet

Inspection Point: Router R0

Header Field	Inbound Frame	Outbound Frame	
Source MAC	PC0 MAC	[R0 (G0/1) MAC]	Changed
Dest MAC	R0 (G0/0) MAC	[R1 (G0/1) MAC]	Changed
Source IP	192.168.1.10	192.168.1.10	
Dest IP	8.8.8.100	8.8.8.100	
TTL	TTL	[TTL - 1]	Decrement

The IP Packet is end-to-end. The Ethernet Frame is hop-by-hop.

The Forwarding Decision Engine



Step 1: Dest IP is 8.8.8.100. Router checks Routing Table.

```
> show ip route  
8.8.8.0/24 via 10.0.0.2
```



Step 2: Router needs MAC for 10.0.0.2. Checks ARP Cache.

```
> show arp  
10.0.0.2 = [R1's Hardware Addr] on GigabitEthernet0/1
```



Step 3: Switch receives frame, checks MAC Address Table.

```
> show mac address-table  
Forward out specific port (not all ports)
```

Synthesis Core Box

Routing identifies the **path**. **ARP** provides the **physical address**. The **MAC table** isolates the **collision domain**. They operate as one **unified machine**.

The Wiretap: Plaintext vs. Encrypted Payloads

HTTP (Cleartext)

```
Source IP: 192.168.1.10
Dest IP: 8.8.8.100

TCP Ports: 80
URL Path: /
Headers:
- Host: www.example.com

Body: <html text>
```

HTTPS (Encrypted)

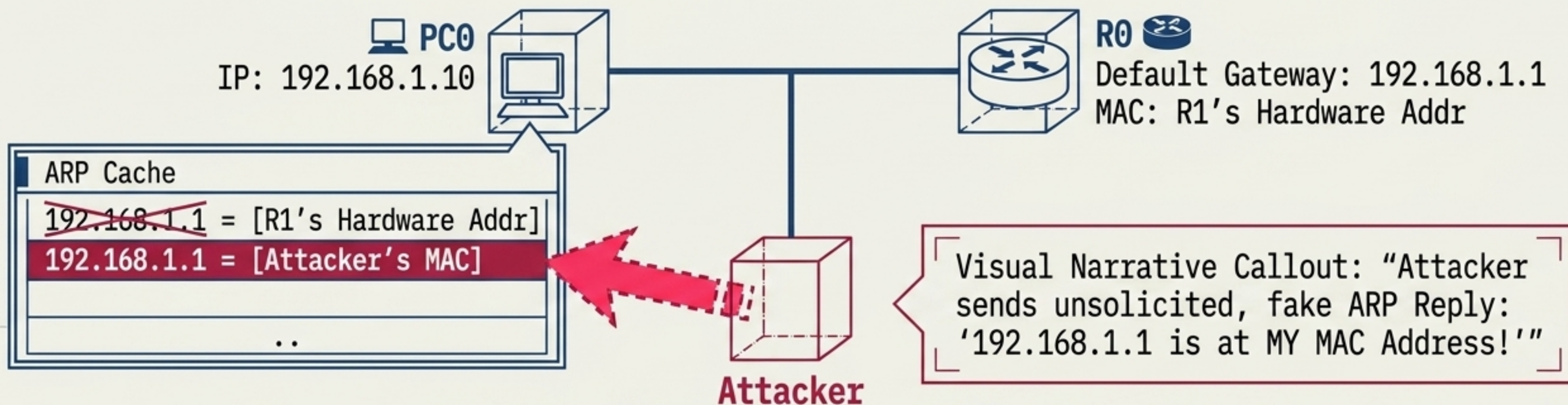
```
Source IP: 192.168.1.10
Dest IP: 8.8.8.100

TCP Ports: 443
[Redacted]
[Encrypted Path]
[Redacted]
[Encrypted Headers]
[Redacted]
[Encrypted Body]
```

Synthesis Core Box

Encryption occurs at the **Application Layer (TLS)**. **Lower layers (IP/TCP)** must remain readable in **plaintext** for the network to route the packet.

The Vulnerability of Trust: ARP Poisoning



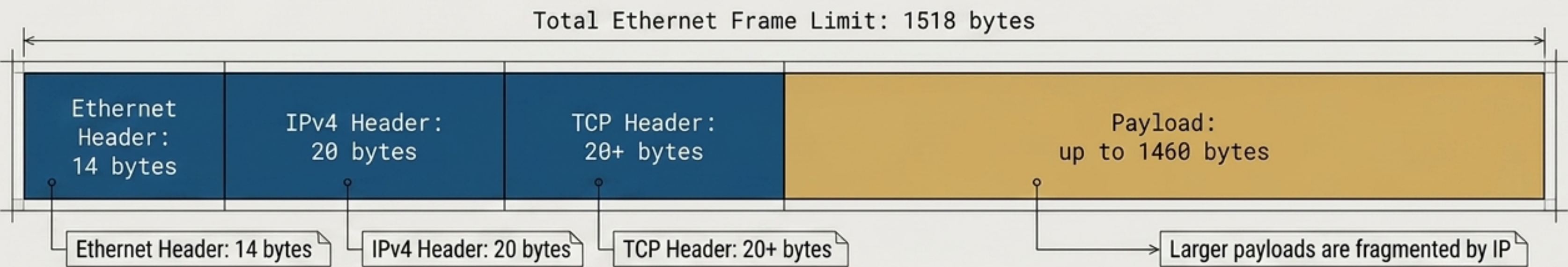
The Exploit Box

PC0's internal ARP table updates instantly. All outbound web requests (DNS, HTTP) are now sent directly to the attacker instead of the legitimate default gateway.

The Root Cause Box

RFC 826 (ARP) includes zero authentication mechanisms. Devices blindly trust any incoming ARP reply, creating a massive local attack surface.

Blueprint Specifications & Terminology



Layer Terminology & Packet Tracer Reference

Layer	Packet Tracer	Concept
Layer 2	Ethernet II frame	Frame
Layer 3	IP datagram	Packet
Layer 4	TCP segment	Segment
Layer 7	HTTP data	Data

Theory defines the rules.
The packet trace proves them.